

A Prototype Study on External Human Machine Interface (HMI) for Automated Bus - Preliminary Testing for Effectiveness Verification Using Virtual Reality (VR)

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ABSTRACT

Autonomous vehicles face safety challenges due to a lack of clear communication with other road users, leading to conflicts, especially at un-signalized intersections. While much research focused on developing communication tools solely for vulnerable road users, this study focuses on developing external Human Machine Interfaces (eHMI) for comprehensive communications in real world. We developed and installed a real electronic LED display board on an autonomous bus to communicate the bus's intention with other road users. To expedite assessment, we created two virtual reality environments—one for optimizing prototype eHMI settings (e.g., its size, positioning, text, etc.) and another for quantitatively evaluating its effectiveness in deadlock situations. These environments complement the gradual accumulation of real-world interaction data. In the research demo, we will demonstrate a real-world new-accident interaction between an autonomous bus and a traditional vehicle, and visitors can experience the reproduced situation in VR with and without the prototype eHMI.

Keywords: HMI, eHMI, Automated Bus, MaaS

Index Terms: Human-centered computing~Human computer interaction (HCI)~Empirical studies in HCI

1 INTRODUCTION

Japan is currently facing various transportation challenges, partly due to its aging population and declining birthrate. One example is the decrease in the workforce for regional public transportation, such as bus and taxi drivers, resulting in an increase in areas without adequate transportation services. The development of autonomous driving technology is a key element in solving these issues, and research in this area is being conducted not only in Japan but around the world.

For the widespread adoption of autonomous vehicles, several challenges remain, including improving safety from a technical

standpoint and developing an understanding of the technology and know-how for service operation in society. One of the problems associated with unmanned autonomous vehicles is the difficulty in conveying the vehicle's actions and intended route to nearby pedestrians and cars, due to the absence of a human driver.

One solution to this problem is the use of a Human Machine Interface (HMI) that mediates between autonomous vehicles and other traffic participants, known as external HMI (eHMI). This replaces the traditional communication methods used by drivers, such as eye contact and gestures, with a system inherent in autonomous vehicles. This research proposes and tests solutions using eHMI for issues identified in the operation of autonomous bus services, aiming to enhance safety and efficiency.

2 RELATED WORKS

As a part of previous studies on eHMI, Fass et al.[1] proposed five types of light-based systems that interact with pedestrians, revealing a high demand among pedestrians to understand the status and intentions of autonomous vehicles.

SIP 2 Report[2] investigated the negative impacts of communication with pedestrians via eHMI for autonomous vehicles. It was reported that when autonomous vehicles prompt pedestrians crossing the street to proceed first, pedestrians tend to neglect checking for oncoming traffic.

3 THEORY

As a model for autonomous buses, we selected the Hitachi BRT, a bus route currently operating in Hitachi City, Ibaraki Prefecture, Japan. Hitachi BRT has been conducting demonstration experiments aimed at realizing a Level 4 autonomous bus service. This research is being carried out with the cooperation of that project with level 2 automation in the real world. The bus route has a dedicated lane for buses, making the introduction of autonomous driving technology relatively easy due to the absence of other vehicles in the lane. This also facilitates the replication of the route in a VR environment, which is part of this paper.

One of the challenges observed in the demonstration experiments of the autonomous bus service on the Hitachi BRT was a deadlock situation at intersections with general roads, where it was unclear when the unmanned bus would start moving upon encountering a passenger car. This report proposes the use of a visual-based eHMI that communicates through text display to solve this issue, and studies its development and evaluation methods. For the eHMI,

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although auditory-based HMIs such as voice broadcasts and calls from remote monitors were considered, this time we primarily focused on a visual-based approach. Auditory-based HMI will be an item for future consideration. Collecting data for this HMI in reality would require using public roads where buses are already operating, necessitating various applications and considerations for nearby residents. Additionally, due to traffic volume and bus operation intervals, there is a risk that enough relevant case samples may not be obtained. On the other hand, VR environments alone may lack realism, necessitating actual verifications in reality as well. Therefore, this study proposes the creation of two types of VR environments for the development and data collection of the eHMI. The first is a VR environment that replicates the actual bus route, allowing for consideration of the placement and design of the HMI as it would appear on the bus. The second is a VR environment that extracts only the traffic environment of the bus route and recreates it in a fictional town. This VR environment enables participants to have a continuous driving experience for about 10 minutes, facilitating the testing of the effects of different parameters, such as HMI display content and duration.

4 DISCUSSION

4.1 Real-world Replication

First, within the VR environment that replicates the actual bus route, we created an event where the subject's car encounters an autonomous bus at an intersection. (Figure 1) Although the experiment was conducted with a small number of participants, it helped us determine a tentative prototype for the HMI that can be installed on actual buses.

This HMI is an LED display board that shows text against a black background, with the text in white and lines above and below the text in yellow. It aims to facilitate communication with the surroundings by displaying relatively short Japanese sentences, about six characters in length.

As shown in the figure, the display board is mounted at the bottom right and left of the bus's front windshield. This eHMI operates at intersections where dedicated bus lanes and general roads meet. Currently, this device has been confirmed to function in demonstration experiments.

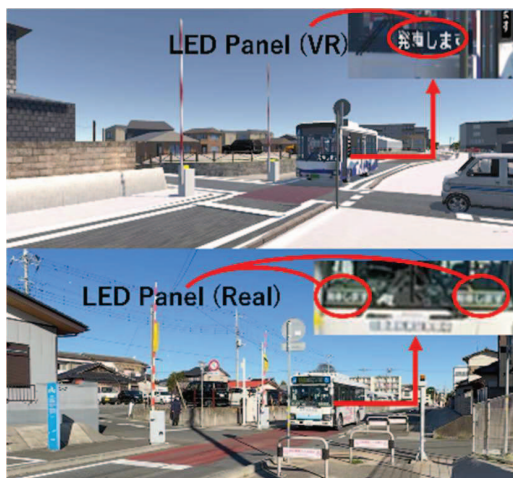


Figure1: Experimental scene (top VR, bottom real world)

4.2 Evaluation of eHMI

The section will describe the evaluation experiment of the developed eHMI. In the second VR environment, participants

encounter multiple events where they meet an autonomous bus at intersections that replicate real-world environments, while having a natural driving experience. (Figure 2) During these encounter events, variables such as the display content of the eHMI, decorative aspects like flashing, and synchronization with the bus's movement enable the collection of subjective evaluations from the participants, such as psychological assessments, as well as objective evaluations like acceleration and deceleration actions. In preliminary experiments conducted by a few subjects, it was confirmed that, in addition to the aforementioned variables, it is also possible to test variables related to road structure, such as the type of traffic signs, the position of stop lines, and which – the bus or the car – has the right of way.

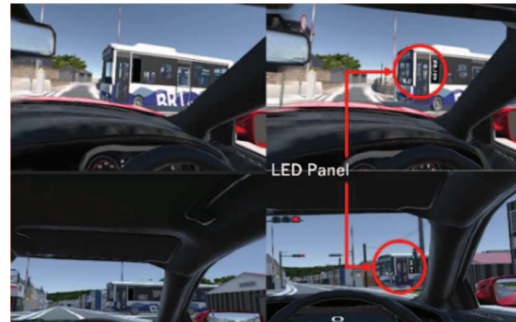


Figure2: Top VR1, Bottom VR2, Left HMI off, Right HMI on.

5 CONCLUSION

Because this paper is a research demonstration, it is difficult to explain all material. In future we will explain these VR experimental environments to verify the acceptance of the eHMI by surrounding traffic participants in situations that are difficult to replicate on the actual bus route (Hitachi BRT) in detail. Additionally, the use of the eHMI in scenarios other than intersections will be considered. At the presentation venue, these two VR environments will be set up, allowing for a brief hands-on experience.

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